



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ELECTRONIC COMMUNICATION				
Type of Course	Minor (GROUP III: SEMICONDUCTOR PHYSICS)				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Fundamentals of EM wave characteristics and electronics				
Course Summary	This course explores the characteristics of the EM wave spectrum, various communication systems and there implementation.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain main parts and different types of electronic communication system. Define electromagnetic spectrum and its application in communication systems.	U & Ap	P	Instructor-created exams / Home Assignments
CO2	Calculate voltage gain, current gain, attenuation. Explain relation between Q, resonant frequency and bandwidth.	Ap	P	Instructor-created exams / Home Assignments
CO3	Explain the basic concepts of AM and FM. Compare AM and FM and calculate parameters such as modulation index, band width.	U & An	P	Instructor-created exams / Home Assignments
CO4	Explain the fundamental concepts in digital communication such as		C	Instructor-created exams / Home

	quantizing error, analog to digital conversion, sampling, PAM, PWM, PPM, difference between asynchronous and synchronous data transmission.	U		Assignments
CO5	Explain the reasons for the growing use of microwaves and millimetre waves in communications. Identify the microwave and millimetre-wave band segments and various microwave components used in this communication system.	U & An	P	Seminar Presentation / Group Tutorial Work
CO6	Design and construct various circuit elements useful in communication systems. Design experiments to identify different characteristics of electromagnetic spectrum.	Ap	P	Practical Assignment / Observation of Practical Skills / Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	INTRODUCTION TO COMMUNICATION SYSTEM		13	20
	1	The significance of human communication, communication system, Types of communication systems.	2	
	2	Modulation and multiplexing, the electromagnetic spectrum	2	
	3	Bandwidth, survey of communication application	2	
	4	Gain, Tuned Circuits	3	
	5	Filters: Passive RC filters, Active filters (advantages, qualitative discussion on op-amp based active filters using circuit diagrams)	2	
	6	Fourier theory	2	
	Relevant topics of chapter 2 of Book 1; sections 1.1-1.7, 2.1, 2.2, 2.3 (selected topics), 2.4 of chapter 2 of Book 1			

II	AMPLITUDE AND FREQUENCY MODULATION		12	18
	7	AM modulation concepts, Modulation index and percentage of modulation	2	
	8	Sideband and frequency domain, pulse modulation	2	
	9	AM power, Single sideband modulation	2	
	10	Basic principles of frequency modulation, principles of phase modulation	2	
	11	Modulation index and side bands, Bessel functions	2	
	12	Frequency suppression effect of FM, AM versus FM	2	
	Relevant topics of chapter 3 and 5 of Book 1; Sections: 3.1 to 3.5, 5.1 to 5.5 of chapter 3 and chapter 5 of Book 1			
III	DIGITAL COMMUNICATION		10	16
	13	Digital transmission of data, serial and parallel transmission	2	
	14	Data conversion, Basic principles of data conversion, General discussion on DA converters and AD converters	2	
	15	Pulse modulation, pulse code modulation	2	
	16	Digital signal processing	2	
	17	Principles of digital transmission	2	
	Relevant topics of chapter 7 and 11 of Book 1; Sections: 7.1 to 7.5, 11.1, 11.2 of chapter 7 and chapter 11 of Book 1			
IV	MICROMETRE AND MILLIMETRE COMMUNICATION		10	16
	18	Microwave concepts, microwave frequencies and band, advantages and disadvantages of microwave transmission, microwave communication system.	2	
	19	Microwave lines and devices	2	
	20	Microwave semiconductor diode	2	
	21	Microwave tubes	2	

	22	Microwave antenna: Low frequency antenna, horn antenna, Microwave and millimetre wave applications	2	
	Relevant topics of chapter 16 of Book 1; Sections: 16.1 to 16.5, 16.7 of chapter 16 of Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course. Necessary theory of experiments can be given as Assignment/ Seminar.			
	1	Design and construct passive RC filters Measure the frequency responses of low-pass and high-pass RC circuits and plot frequency response graphs (Bode plots) of the amplitude and the phase.		
	2	Construct amplitude modulator circuit - Design and construct an amplitude modulator circuit. - Study the response for suitable modulation depths.		
	3	Construction of D/A converter - Construct a 4 bit D/A converter using R-2R ladder network. - Plot a graph of analog output voltage versus binary number.		
	4	Determine the numerical aperture (NA) of an optical fiber using a laser - Couple the light from the laser source onto one of the fiber ends and the light coming from the other end is allowed to fall on a screen(sheet having circular markings) placed perpendicular to the axis of the fiber. - Measure the diameter of the laser beam on the screen and the distance between the screen and fiber output end and hence calculate the NA.		

5	Attenuation and bandwidth of optical fibre Determine the attenuation and bandwidth of the given optical fibre specimen		
6	Fourier analysis of the modes of vibration in a stretched string. - Record the sound produced by guitar string (or similar arrangement) using a microphone and analyze the spectrum by taking FFT. - Audio Spectrum in the Pyphox, Audacity, ExpEYES or any other tools can be used to record the sound and to take FFT. - Vary the length and tension of the string and analyze the harmonics. https://phyphox.org/experiment/audio-spectrum/ https://www.youtube.com/watch?v=bl7jf2myEvM https://expeyes.in/experiments/sound/beats.html		
7	Construct Half adder using universal gates and study the operation. Implement half adder using NAND/NOR gates and verify the truth table for each input/output combination.		
8	Verification of De-Morgan's Theorems using basic gates. Realize the either side of the De-Morgan's Theorems using gates from appropriate ICs and verify the truth table for each input/output combination.		
9	Construct and study the operations of the RS and JK Flip-Flops using IC's - Realize RS Flip-Flop using NAND gates and verify the truth table - Realize JK Flip-Flop using NAND gates from appropriate ICs and verify the truth table		
10	Construction of the center tapped full wave rectifiers and regulated power supply. - Construct a center tapped full wave rectifier without filter and with a filter. - Measure the AC and DC voltages using a multimeter and calculate the ripple factor without and with a filter.		

		- Observe the variation of the ripple factor with load resistance, when filter is used. - Construct 5V/12V regulated power supply using 78XX IC.		
	11	Study the frequency response of common emitter(CE) transistor amplifier. - Design a CE transistor amplifier of a given gain (mid-gain) using voltage divider bias. - Analyse the frequency response, draw the curve and find the bandwidth, without feedback.		
	12	Construction of LC oscillator (Hartley or Colpitt's) Construct a LC oscillator (Hartley or Colpitt's) and measure the frequency using CRO/ExpEYES for different values of L and C. Compare with the theoretical values.		
	13	Determination of Plank's constant using LEDs - Observe the turn-on voltage, V_0 of LEDs and calculate the value of h. Use at least 4 different colors of LED (with transparent casing) - Plot $\frac{1}{\lambda} - V_0$ graph using Python, fit a straight line to get the slope and estimate the value of h. - Calculate the %error. - Programmable voltage source of ExpEYES may be used to find the turn-on voltage.		
	14	Analysis of Hydrogen spectra using the Tracker Video Analysis tool. - Calibrate the video of the Hydrogen spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra. - Plot the intensity profile, find the prominent wavelengths of the Balmer series and calculate the Rydberg's constant. - Estimate the %error. - Pre recorded video of the Hydrogen spectra can be used. https://physlets.org/tracker/. https://www.youtube.com/watch?v=UCCPkJpUQEw		

	15	Black body spectrum of Sun -Estimation of surface temperature using the Tracker Video Analysis tool. - Calibrate the video of the solar spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra. - Plot wavelength vs intensity, get λ_{max} and using Wein's law calculate the surface temperature. - Pre recorded video of the solar spectra can be used.		
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Books and References:

- Principles of electronic communication system, 4th Edition by Louis E. Frenzel (Book 1)
- Electronic communication systems, 5th Edition by y George Kennedy, Brendan Davis, Srm Prasanna- Mc-Graw Hill(Book 2)
- Electronic Communications System, 5th Edition by Wayne Tomasi, Pearson (Book 3)
- Principles of Electronics, 11th edition by V.K. Mehta and Rohith Mehta, S Chand & Company (Book 4)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	1	0	0	2	3	3	3	2	1	1	0
CO 2	3	2	1	0	0	2	3	3	3	2	1	1	0
CO 3	3	2	1	0	0	2	3	3	3	2	1	1	0
CO 4	3	2	1	0	0	2	3	3	3	2	1	1	0
CO 5	3	2	1	0	0	2	3	3	3	2	1	1	0
CO 6	3	2	2	0	0	3	3	3	3	2	1	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Practical Skill Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	